Methodology and Data Integration

Semester: DA Semester 4

Course: Capstone 2

1. Project Overview, Context, and Measurable Objectives

Healthcare costs in the United States are heavily influenced by a combination of hospital operations, financial decisions, labor market pressures, geographic context, and local socioeconomic conditions. Traditional hospital cost studies frequently analyze these factors in isolation.

This capstone project builds upon prior work completed in an earlier semester, where large-scale public hospital cost report data was collected, cleaned, explored, and visualized. That initial phase established a validated analytical dataset and produced dashboards summarizing financial and operational trends. The current semester advances the project from descriptive analytics to predictive and decision-oriented analytics, with a distinct focus on machine learning-based regression and classification.

The emphasis of our modeling is on analyzing institutional cost efficiency, categorizing performance risk, and interpreting results in a manner that mirrors real-world analytical decision support. Hospitals operate under fixed reimbursement structures while attempting to manage rising operational costs. In practice, healthcare organizations rely on two complementary types of analytics: numeric forecasts for planning and budgeting, and categorical risk flags for prioritization and intervention. This capstone addresses both needs by harmonizing regression and classification within a single modeling framework.

The measurable objectives of this capstone are explicitly defined:

- Prepare a real-world, multi-year institutional dataset for machine learning.
- Predict hospital cost efficiency using supervised regression.
- Convert continuous efficiency measures into actionable classification categories.
- Build and compare baseline and advanced machine learning models.
- Perform rigorous evaluation and error analysis.
- Interpret model outputs to identify drivers of inefficiency and risk.
- Demonstrate the complete data science lifecycle in a capstone-level project.

2. High-Level Data Architecture and Repository Structure

To achieve our analytical goals, the project integrates four major data sources, each functioning at a distinct geographic and analytical level:

- **CMS Hospital Cost Reports:** Serves as the core analytical dataset at the Hospital-Year level.
- **RUCC Codes:** Provides Rural-Urban classification at the County level.
- **BLS OEWS Data:** Measures workforce wage pressure at the State-Year-Occupation level.
- **Census ACS (5-year):** Captures socioeconomic context at the County-Year level.

The final machine learning dataset rigorously aligns all data at the hospital-year level. To maintain a reproducible data pipeline, files are organized into a strict folder hierarchy.

Supplementary Data Folder (Raw and Intermediate Products):

- Census files: census_raw_2012_2022.csv, census_county_renamed_2012_2022.csv, census_ml.csv.
- BLS files: bls_all_years_combined.csv, bls_all_occupations_standardized.csv, bls_healthcare_all_occupations_data.csv, bls_healthcare_grouped.csv, bls_hospital_wage_group_dataset.csv.
- Associated notebooks and raw sets: BLS_Healthcare_Data.ipynb, BLS Datasets/, and Old Files.

ML Modelling Folder (Final Integrations and Outputs):

- Stepwise integration notebooks: combined_bls_files.ipynb, Aggregated_BLS.ipynb, Census_data.ipynb, CMS_Urban_Rural_Dataset.ipynb, CMS_Urban_Rural_BLS_Dataset.ipynb, CMS_BLS_Census.ipynb.
- Compiled datasets: hospital_kpi_ready_with_rucc.csv, hospital_kpi_with_bls_wages.csv, hospital_kpi_cms_census_rucc.csv, hospital_cms_census_bls_rucc.csv.

3. Completed Phase 1: Granular Data Engineering Pipeline

The objective of our data engineering pipeline is to construct a comprehensive, machine-learning-ready dataset. We achieved this through the following steps.

3.1. Base Dataset Formulation: CMS Hospital Cost Data

The Centers for Medicare & Medicaid Services (CMS) data forms the core dataset and defines the row-level granularity of the project (one row per hospital per year spanning 2011 to 2022). This base establishes the dependent variables and core predictors for downstream machine learning models.

- **Capacity and Utilization Features:** Number of beds, bed-days available, occupancy rate, total discharges, and length of stay.
- **Workforce Features:** Full-time equivalent employees.
- **Financial Features:** Revenues, costs, assets, liabilities, and derived ratios including profit margin, cost-to-charge ratio, and revenue per bed.
- **Preprocessing:** The dataset was cleaned to ensure consistent numeric types, removal of redundant fields, and the creation of interpretable derived metrics representing operational efficiency, financial health, and care intensity.

3.2. Geographic Classification Integration (RUCC)

To enable direct rural vs urban comparisons, provide policy-relevant interpretation, and control for geographic healthcare access differences, we integrated the USDA Census-aligned Rural-Urban Continuum Codes (RUCC).

- **The Challenge:** CMS data does not provide county FIPS codes, and RUCC data is indexed by county name and state.
- **The Engineering Approach:** We implemented a robust county name normalization process in both datasets. This involved uppercasing text, removing punctuation, handling special geographic cases (like Alaska boroughs), and removing suffixes such as County, Parish, Municipio, Borough, and Census Area.
- **The Output:** RUCC values were filtered to RUCC_2023 and transformed into a numeric rucc_code and a binary rural_urban classification, where hospitals are flagged as Urban if their RUCC is $\le$ 3, and Rural otherwise. This output became the base dataset for all further merges. Missingness was controlled and expected (~8–11%) due to unmatched county names, small/special jurisdictions, or territorial records.

3.3. Healthcare Labor Market Wage Integration (BLS)

The BLS Occupational Employment and Wage Statistics (OEWS) data captures healthcare labor market conditions, which are a major driver of hospital operating costs.

- **Standardizing a Unified Master Table:** A major challenge was that BLS file schemas vary significantly by year; column names, casing, and formats change, with early years using .xls and later years using .xlsx. All state-level files from 2011 to 2022 were read using version-appropriate Excel engines and merged into a single dataset (bls_all_years_combined.csv). We preserved all original columns (68+) to serve as an audit table ensuring no information loss.

- **Isolating Healthcare Occupations:** We utilized SOC codes to extract Healthcare Practitioners & Technical Occupations (29-xxxx) and Healthcare Support Occupations (31-xxxx).

- **Occupation Grouping Strategy:** To balance interpretability and dimensionality, roles were grouped into five categories: Physician (highest wage, strongest cost impact), Practitioner (non-physician clinicians), RN (critical staffing group), Support (frontline and auxiliary staff), and Other (remaining healthcare roles). Occupation titles were explicitly retained to preserve semantic meaning.

- **Wage Aggregation Methodology:** Simple averaging of wages would distort labor cost signals because occupations differ greatly in employment size. The group successfully implemented an employment-weighted average wage formula: $\text{Average Wage} = \frac{\sum (a\_mean \times tot\_emp)}{\sum (tot\_emp)}$. This accurately reflects labor market cost pressure and the true economic exposure faced by hospitals.

- **Dataset Explosion Diagnosis:** An initial attempt to merge occupation-level BLS data directly into CMS caused the dataset to expand beyond 4 GB due to a many-to-many join. This issue was successfully resolved by aggregating BLS wages *before* merging, and merging only state-year wage features into the CMS file to avoid row explosion. Multiple BLS tables were created, culminating in wide-format ML wage features (e.g., wage_physician, wage_rn).

3.4. Community Context and Socioeconomic Integration (Census)

Census ACS 5-Year Estimates were retrieved using the U.S. Census Bureau API at the county-year level spanning 2012 to 2022. This data provides socioeconomic and demographic context essential for interpreting hospital performance beyond internal operations.

- **Variables Extracted:** population (County population), median_age (Demographic structure), median_income (Economic capacity), poverty_rate (Economic stress), and higher_education_rate (Workforce quality proxy).

- **Missing Data Analysis:** A year-wise missingness analysis revealed that 2011 was 100% missing because no ACS 5-year data was available structurally. Data from 2012 to 2022 showed ~6–11% missingness due to suppressed estimates, small counties, and geographic boundary changes.

This missingness is structural and expected, confirming the correctness of our API calls and merge logic. Census data was merged at the county-year level using FIPS codes, then aligned to hospitals via county mapping.

3.5. Final Integrated Dataset Output

The data engineering pipeline finalized in the hospital_cms_census_bls_rucc.csv dataset. It currently holds approximately 75,870 rows and 56 columns, with a granularity of one row per hospital per year. It features comprehensive coverage of hospital operations, rural-urban classifications, state-level labor costs, and county-level socioeconomic indicators. The dataset passes ML readiness assessments with controlled missingness, interpretable feature design, numeric features available, no many-to-many joins, and no unintended row duplication.

4. Planned Phase 2: Machine Learning Problem Decomposition

With data engineering complete, the project transitions to three interrelated machine learning tasks, each serving a distinct analytical purpose.

4.1. Feature Engineering Strategy and Leakage Control

Feature engineering will extend beyond raw variable usage.

- **Financial Features:** We will convert raw net patient revenue and total operating expenses into engineered features such as expense ratios, revenue per bed, and cost intensity indicators to normalize scale differences across institutions. We will also create financial stability ratios like debt-to-asset ratios.

- **Operational Features:** Raw total beds, average daily census, and full-time equivalent staffing will be engineered into occupancy rates, staff-to-bed ratios, and utilization-adjusted staffing to capture true efficiency rather than size.

- **Structural and Temporal Features:** We will apply one-hot encoding for hospital type, ownership, geographic aggregation, and reporting year to capture temporal effects.

- **Data Leakage Prevention:** To ensure methodological rigor, variables directly used to compute targets (e.g., removing the cost of charity care when predicting the charity care ratio) will be excluded from predictors to prevent target leakage and artificially inflated model performance. We will also handle missing values using feature-appropriate imputation, treat outliers using robust statistical techniques, and use train-test splits that preserve temporal integrity.

4.2. Task 1: Regression (Primary Task)

Our objective is to train multiple regression models to predict the Cost-to-Charge Ratio (CCR) for each hospital-year observation, as well as the hospital Profit Margin.

- **Mathematical Definition:** $\text{CCR} = \frac{\text{Total Operating Costs}}{\text{Total Charges}}$.
- **Rationale:** CCR is chosen because it is a direct measure of cost efficiency, is consistently available across reporting years, is less volatile than profit margin, and is commonly used in institutional benchmarking and reimbursement analysis. This answers the question: "How efficient is this hospital expected to be?".
- **Modeling Strategy:** We will introduce models progressively. Explainable baselines will include Linear Regression, Ridge Regression, and Lasso Regression. K-Nearest Neighbors (KNN) will also be utilized to predict outcomes based on similarity between hospitals (e.g., matching facilities with similar bed capacity and workforce structures). Advanced, performance-focused models will include the Random Forest Regressor and Gradient Boosting Regressor.

4.3. Task 2: Classification (Decision-Support Task)

Our objective is to classify hospitals into efficiency tiers and risk categories based on CCR. Classification labels will not be arbitrary; they are derived directly from the continuous CCR distribution to ensure objectivity and reproducibility.

- **Multi-Class Efficiency Tier Classification:** Hospitals are grouped using quantile-based thresholds derived from the continuous CCR distribution: the lowest 25% of CCR values are labeled "High Efficiency", the middle 50% are labeled "Medium Efficiency", and the highest 25% are labeled "Low Efficiency". This avoids subjective cutoffs and adapts automatically to data distribution, aligning with peer comparison use cases.
- **Binary Risk Flag Classification:** A binary risk indicator maps an "At-Risk" definition to CCR values above a high percentile (e.g., 75th or 90th), and "Not At-Risk" to CCR below the threshold. This supports early warning, prioritization, simplifies intervention decisions, and mirrors real-world screening workflows.
- **Modeling Strategy:** The classification models will learn decision boundaries that map financial, operational, and structural features to these efficiency or risk categories. Baselines will utilize Logistic Regression, while advanced strategies will utilize Random Forest Classifier, Gradient Boosting Classifier, and Decision Trees that create rule-based splits based on occupancy and financial ratios. Class imbalance will be systematically addressed using class weighting and stratified sampling.

4.4. Task 3: Unsupervised Learning (Supporting Task)

While unsupervised learning is not our primary outcome, we will use it to cluster hospitals into peer groups based on similarity in operational and financial profiles. Since there is no target variable, clustering hospitals by financial or operational similarity will provide context for regression errors, improve the interpretation of classification outcomes, and enable peer-based benchmarking.

5. Evaluation, Error Analysis, and Interpretability Generation

We will evaluate models using statistical performance metrics and interpretability techniques.

- **Regression Evaluation:** Performance will be quantified using the R^{2} Score (to measure variation explained by the model), Mean Squared Error (MSE, capturing average squared prediction error), and Root Mean Squared Error (RMSE, providing easier interpretation of prediction error). We will also review error distribution by hospital type and region.

- **Classification Evaluation:** Categorical models will be assessed via Accuracy, Precision, Recall, F1-score, and Confusion matrices.

- **Diagnostic Analysis:** We will monitor for underfitting (where a model like Linear Regression is too simple to capture complex nonlinear patterns) and overfitting (where a model like a lone Decision Tree memorizes training data but performs poorly on new test data). We expect Random Forest to act as our balanced model by combining multiple decision trees to capture complex patterns while maintaining generalization. Diagnostic processes will identify systematic prediction errors, compare baseline vs advanced model failures, and ensure the stability of classification across years.

- **Interpretability and Finding Generation:** Interpretation methods will include coefficient analysis for linear models, comparisons of drivers across efficiency tiers, and explicitly utilizing Permutation Feature Importance. Permutation Feature Importance measures how model accuracy changes when feature values are shuffled, identifying key drivers of inefficiency and evidence-based improvement levers. We will utilize this to analyze relationships, such as how profit margin is strongly influenced by internal hospital metrics (revenue generated per bed, debt-to-asset ratio, cost management) and how Charity Care Ratio is strongly influenced by external variables (Medicare CBSA geographic region, community population, socioeconomic conditions). The addition of our Census and BLS wage external datasets explicitly aids in understanding how hospitals are influenced by external population demand factors.

6. Next Steps and Integration with Prior Work

Our planned next steps are to decide the final treatment of the 2011 missing data (whether to drop or impute), finalize our ML target variables, encode categorical variables, create the ML-only feature matrix, and train baseline ML models. Dashboards developed in the previous semester will be reused to validate ML findings visually, provide continuity in storytelling, and support the final presentation and defense. Streamlit dashboards will be utilized for final insights. This pipeline emphasizes correctness, interpretability, and scalability, making it suitable for both academic research and applied machine learning